

Leapfrog Integration Algorithm

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August 5, 2026

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1 Motivation

When solving Newton's equations numerically, we seek an algorithm that

- ▷ is accurate,
- ▷ is computationally inexpensive,
- ▷ does not slowly destroy the physics.

The simplest algorithm, Euler integration, satisfies only the second criterion. It gradually gains or loses energy, making planetary orbits spiral outward or inward after enough time steps.

Leapfrog integration solves this problem with one remarkably simple idea:

Store positions and velocities at different times.

Rather than updating both variables simultaneously, they are shifted by half a time step and take turns updating one another.

2 The Equations of Motion

Suppose a particle obeys

$$\frac{d\mathbf{x}}{dt} = \mathbf{v},$$

and

$$\frac{d\mathbf{v}}{dt} = \mathbf{a}(\mathbf{x}),$$

where

$$\mathbf{a} = \frac{\mathbf{F}}{m}.$$

Our goal is to compute

$$\mathbf{x}(t), \quad \mathbf{v}(t)$$

at discrete time intervals

$$t_n = n\Delta t.$$

3 Why Euler Integration Fails

Euler updates both quantities using information from the current time.

$$\mathbf{v}_{n+1} = \mathbf{v}_n + \mathbf{a}_n \Delta t$$

$$\mathbf{x}_{n+1} = \mathbf{x}_n + \mathbf{v}_n \Delta t.$$

Although simple, this method always extrapolates using old information. The errors accumulate in one direction, producing artificial energy gain or energy loss. Eventually,

- ▷ planets escape,
- ▷ oscillators explode,
- ▷ pendulums slow down,
- ▷ simulated systems drift away from reality.

4 The Central Idea of Leapfrog

Leapfrog stores

Position at integer times

and

Velocity at half-integer times.

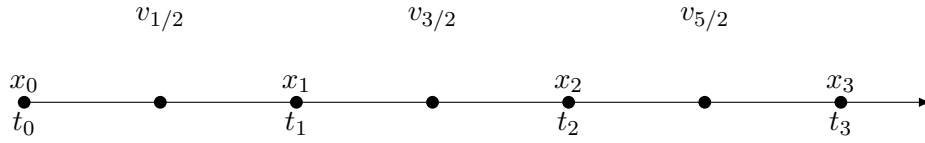
That is,

$$x_0, x_1, x_2, \dots$$

but

$$v_{1/2}, v_{3/2}, v_{5/2}, \dots$$

The variables literally leap over each other.



Notice that every velocity lies exactly halfway between two positions.

5 Why Half-Step Velocities?

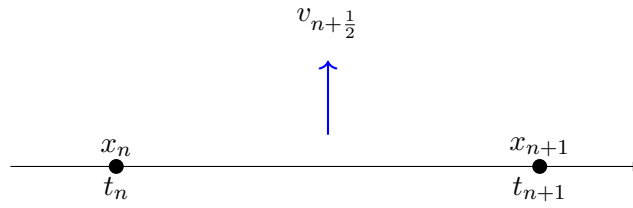
Suppose we approximate

$$\frac{dx}{dt}.$$

Using finite differences,

$$\frac{x_{n+1} - x_n}{\Delta t},$$

this derivative naturally belongs halfway between x_n and x_{n+1} . Visually,



Likewise,

$$\frac{v_{n+\frac{1}{2}} - v_{n-\frac{1}{2}}}{\Delta t}$$

belongs exactly at time t_n , which is where the acceleration is computed. Everything is perfectly aligned.

6 The Leapfrog Algorithm

Suppose we already know

$$x_n, \quad v_{n-\frac{1}{2}}.$$

Step 1

Compute acceleration

$$a_n = a(x_n).$$

Step 2

Advance velocity

$$v_{n+\frac{1}{2}} = v_{n-\frac{1}{2}} + a_n \Delta t.$$

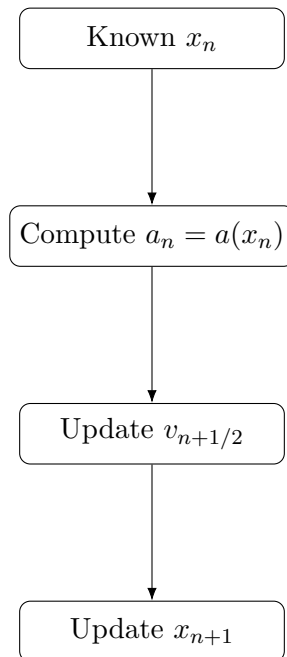
Step 3

Advance position

$$x_{n+1} = x_n + v_{n+\frac{1}{2}}\Delta t.$$

This completes one time step.

7 Flow Diagram



8 Worked Example

Suppose

$$x_0 = 0, \quad v_{1/2} = 1, \quad a = 2, \quad \Delta t = 1.$$

Velocity:

$$v_{3/2} = 1 + 2 = 3.$$

Position:

$$x_1 = 0 + 3 = 3.$$

Next step,

$$v_{5/2} = 3 + 2 = 5,$$

$$x_2 = 3 + 5 = 8.$$

Notice the alternating updates.

9 Comparison with Euler

Property	Euler	Leapfrog
Accuracy	First order	Second order
Energy conservation	Poor	Excellent
Time reversible	No	Yes
Long-term stability	Poor	Excellent
Computational cost	Low	Low

10 Geometric Interpretation

The algorithm may be viewed as two runners on a track.

One runner represents position.

The other represents velocity.

Neither runner ever stands beside the other.

Instead,

$$x \rightarrow v \rightarrow x \rightarrow v \rightarrow x$$

Each update leaps over the previous variable.

Hence the name

Leapfrog Integration.

11 Pseudo-code

```
Given x and v_half
repeat
    a = acceleration(x)
    v_half = v_half + a*dt
    x = x + v_half*dt
end
```

12 Advantages

Leapfrog is popular because it is

- ▷ second-order accurate,
- ▷ symplectic,
- ▷ time reversible,
- ▷ computationally cheap,

- ▷ excellent for long simulations,
- ▷ widely used in celestial mechanics, molecular dynamics, plasma physics, particle simulations, and N -body gravity.

13 Key Takeaways

1. Positions are stored at integer times.
2. Velocities are stored at half-integer times.
3. Velocity is updated first.
4. Position is updated second.
5. The two variables leap over one another.
6. This staggering dramatically improves long-term stability.
7. Leapfrog preserves the geometry of Hamiltonian mechanics, making it one of the most important numerical integrators in computational physics.

$$\begin{aligned} v_{n+\frac{1}{2}} &= v_{n-\frac{1}{2}} + a_n \Delta t, \\ x_{n+1} &= x_n + v_{n+\frac{1}{2}} \Delta t. \end{aligned}$$